

Assignment 1

1) Determine the pole(s) of the vehicle system:

Equation

$$\frac{dV(t)}{dt} = -\frac{a}{m} V(t) + \frac{b}{m} u(t) + d(t)$$

Laplace - transform

$$sV(s) = -\frac{a}{m} V(s) + \frac{b}{m} U(s) + D(s) \Rightarrow$$
$$\Rightarrow sV(s) + \frac{a}{m} V(s) = \frac{b}{m} U(s) + D(s) \Rightarrow$$
$$\Rightarrow V(s) \left(s + \frac{a}{m} \right) = \frac{b}{m} U(s) + D(s)$$

Transfer function:

$$H(s) = \frac{V(s)}{U(s)} = \frac{\frac{b}{m}}{s + \frac{a}{m}}$$

Pole(s)

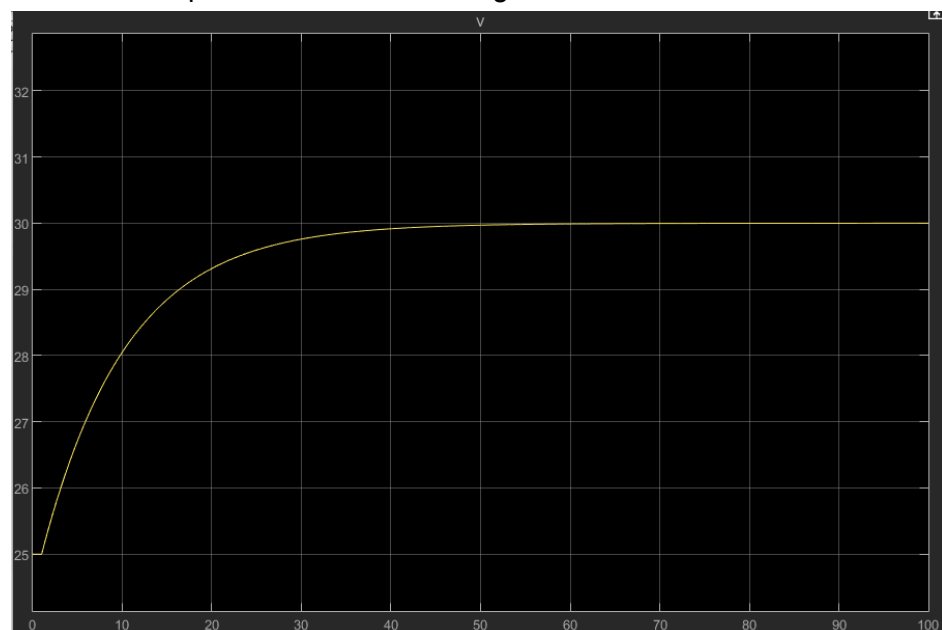
$$s + \frac{a}{m} = 0 \Rightarrow s = -\frac{a}{m}$$

1 Pole $\Rightarrow s = -\frac{a}{m}$

1 pole $\Rightarrow s = -a/m = -0.104712041884817$

2) Implement the model equation in Simulink for simulation analysis:

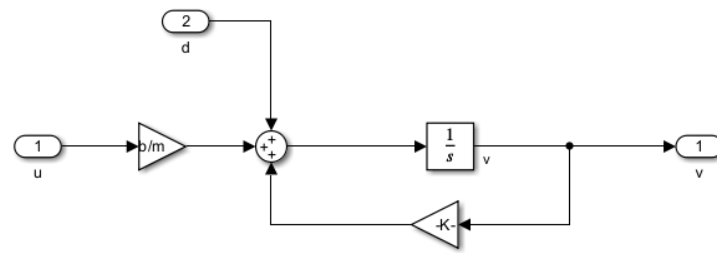
Photo of the speedcurve when running the model:



The steady-state speed is equal to 30 m/s.

Initial speed is equal to 25 m/s.

How our "Vehicle mode" looked in simulink:



3) To determine the characteristics equation for the closed-loop system:

$$\begin{aligned}
 G(s) &= \frac{b}{s + \frac{a}{m}} & F(s) &= K_p + \frac{K_i}{s} \\
 \text{open loop transfer function} \\
 L(s) &= G(s) \cdot F(s) = \left(\frac{b}{s + \frac{a}{m}} \right) \cdot \left(K_p + \frac{K_i}{s} \right) \Rightarrow \\
 \Rightarrow L(s) &= \frac{\frac{b}{m} (K_p s + K_i)}{s(s + \frac{a}{m})} \\
 \text{closed loop transfer function} \\
 T(s) &= \frac{L(s)}{1 + L(s)} \\
 T(s) &= \frac{\frac{b}{m} (K_p s + K_i)}{s(s + \frac{a}{m}) + \frac{b}{m} (K_p s + K_i)} \Rightarrow \\
 \Rightarrow T(s) &= \frac{\frac{b}{m} (K_p s + K_i)}{s(s + \frac{a}{m}) + \frac{b}{m} (K_p s + K_i)} \\
 \hookrightarrow s(s + \frac{a}{m}) + \frac{b}{m} (K_p s + K_i) &= 0 \Rightarrow \\
 \Rightarrow s^2 + \left(\frac{a}{m} + \frac{b}{m} K_p \right) s + \frac{b}{m} K_i &= 0
 \end{aligned}$$

The characteristic equation of the closed-loop system is:

$$s^2 + \left(\frac{a}{m} + \frac{b}{m} * k_p \right) * s + \frac{b}{m} * k_i$$

- 4) Study the behavior of the closed-loop system using the controllers designed in a previous problem and the Simulink model:

Calculations:

Problem 4

Case a) -0.2 and -0.8 .

$$(s - p_1)(s - p_2) = 0$$

$$(s + 0.2)(s + 0.8) = 0$$

$$s^2 + s + 0.16$$

characteristic eqn.

$$s^2 + \left(\frac{a}{m} + \frac{b}{m} k_p\right)s + \frac{b}{m} k_i$$

$$\frac{a}{m} + \frac{b}{m} k_p = 1$$

$$k_p = \frac{m-a}{b} \quad k_i = 0.16 \times \frac{m}{b}$$

$$k_p = 0.1541 \quad k_i = 0.0275$$

case b)

$$-0.5 \pm j\sqrt{2}$$

$$[s - (-0.5 + j\sqrt{2})][s - (-0.5 - j\sqrt{2})]$$

$$(s + 0.5 - j\sqrt{2})(s + 0.5 + j\sqrt{2})$$

$$(s + 0.5)^2 - (j\sqrt{2})^2$$

$$(s + 0.5)^2 - (-2) \Rightarrow (s + 0.5)^2 + 2$$

$$s^2 + 2s + 0.5^2 + 2 \Rightarrow s^2 + s + 0.25 + 2$$

$$s^2 + s + 2.25$$

Characteristic Eqn

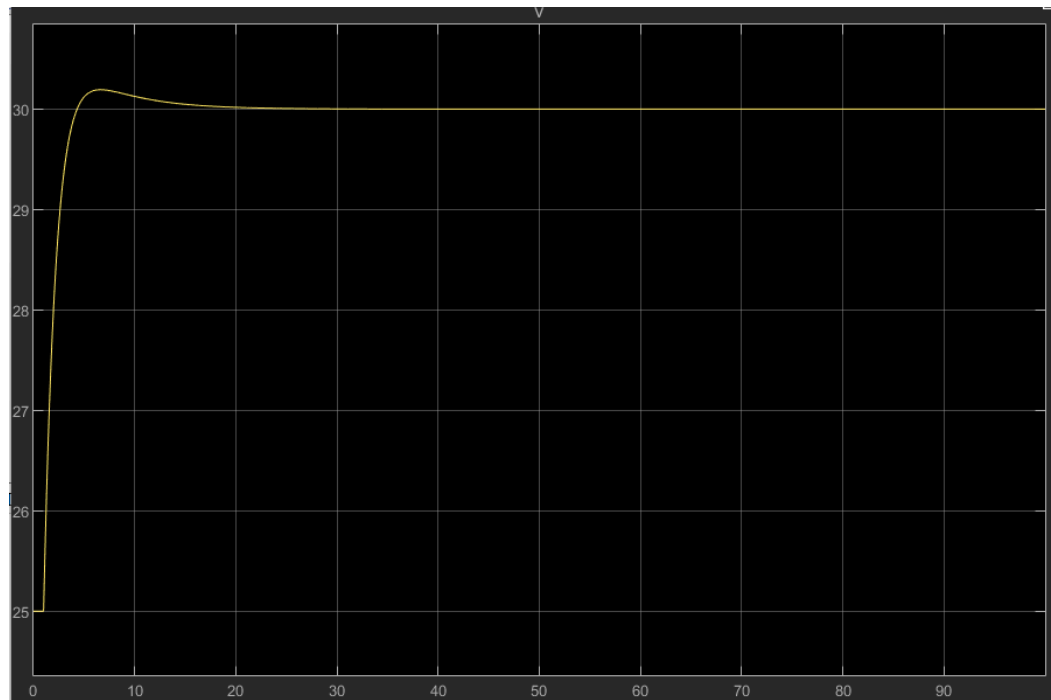
$$s^2 + \left(\frac{a}{m} + \frac{b}{m} k_p\right)s + \frac{b}{m} k_i$$

$$\frac{a}{m} + \frac{b}{m} k_p = 1$$

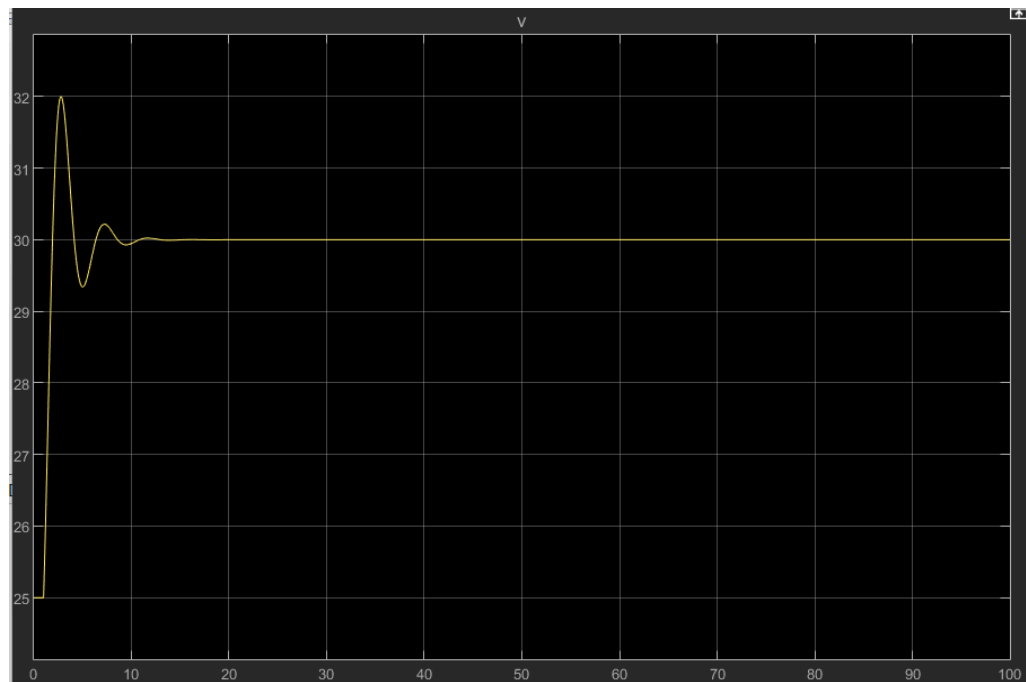
$$k_p = \frac{m-a}{b} \quad k_i = 2.25 \times \frac{m}{b}$$

$$k_p = 0.1541 \quad k_i = 0.3872$$

Case a) -0.2 and -0.8 :



Case b) $-0.5 \pm j\sqrt{2}$



- 5) Determine the rise time, the overshoot, and the settling time for the two controllers:

This questions is answered in matlab and we got the answered:

a) Poles in -0.2 and -0.8:

```
RiseTime_5a = 2.0845;
SettlingTime_5a = 2.6392;
Overshoot_5a = 3.5008;
```

b) Poles in $-0.5 \pm j\sqrt{2}$

```
RiseTime_5b = 0.7111;
SettlingTime_5b = 4.9343;
Overshoot_5b = 40.0275;
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- 6) Given the model above, determine the transfer function $G_{uy}(s)$ from the throttle angle to the relative distance, and the transfer function $G_{wy}(s)$ from the disturbance to the relative distance:

Problem 6 :

The state space Model is

$$\dot{x} = Ax + Bu + Bw$$

$$y = Cx + Du + Dw$$

$$A = \begin{bmatrix} 0 & -1 \\ 0 & -a/m \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ b/m \end{bmatrix} \quad B_w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \end{bmatrix} \quad D_w = \begin{bmatrix} 0 \end{bmatrix}$$

Transfer function for state space.

$$G_{uy}(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & 1 \\ 0 & s + \frac{a}{m} \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} s + \frac{a}{m} & -1 \\ 0 & s \end{bmatrix}$$

$$G_{uy}(s) = (1 \ 0) \times \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} s + \frac{a}{m} & -1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 0 \\ b/m \end{bmatrix} + (0)$$

$$= (1 \ 0) \times \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} 0 & -\frac{b}{m} \\ 0 & s \frac{b}{m} \end{bmatrix}$$

$$= (1 \ 0) \times \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} -\frac{b}{m} \\ s \frac{b}{m} \end{bmatrix}$$
~~$$G_{uy}(s) = C \begin{bmatrix} -\frac{b}{m} \\ s \frac{b}{m} \end{bmatrix} \Rightarrow (1 \ 0) \begin{bmatrix} -\frac{b}{m} \\ s \frac{b}{m} \end{bmatrix} \Rightarrow$$~~

$$G_{uy}(s) = \frac{1}{s(s + \frac{a}{m})} \times \left(-\frac{b}{m}\right) \Rightarrow \frac{-\frac{b}{m}}{s(s + \frac{a}{m})}.$$

$$\boxed{G_{uy}(s) = \frac{-\frac{b}{m}}{s(s + \frac{a}{m})}}$$

$$G_{wy}(s) = C(sI - A)^{-1} B_w$$

$$= (1 \ 0) \times \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} s + \frac{a}{m} & -1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= (1 \ 0) \times \frac{1}{s(s + \frac{a}{m})} \begin{bmatrix} s + \frac{a}{m} & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \frac{1}{s(s + \frac{a}{m})} \left[\cancel{s + \frac{a}{m}} \right] \Rightarrow \frac{1}{s}$$

$$\boxed{G_{wy}(s) = \frac{1}{s}}$$

7) Use pole placement design to design an adaptive cruise controller in the form of a PID controller:

Problem 7:-

$$F(s) = kp + \frac{k_i}{s} + k_d s \quad p = -0.2$$

$$\begin{aligned} (s+0.2)^3 &= s^3 + 0.2^3 + 3 \times s \times 0.2(s+0.2) \\ &= s^3 + 0.008 + 0.6s(s+0.2) \\ &= s^3 + 0.6s^2 + 0.12s + 0.008 \quad \text{--- (1)} \end{aligned}$$

The closed loop transfer function is $1 + F(s)G(s) = 0$.

$$G(s) = \frac{-b/m}{s(s+q/m)} = s$$

$$= 1 + \left(kp + \frac{k_i}{s} + k_d s\right) \left(\frac{-b/m}{s(s+q/m)}\right) = 0$$

$$\Rightarrow G(s)F(s) \Rightarrow \frac{-b/m (kps + k_i + k_d s^2)}{s^2(s+q/m)}$$

$$1 + G(s)F(s) \Rightarrow 1 + \frac{-b/m (kps + k_i + k_d s^2)}{s^2(s+q/m)}$$

$$1 + G(s)F(s) = 0$$

$$s^2(s+q/m) - b/m (kps + k_i + k_d s^2) = 0$$

$$= s^3 + \frac{q}{m} s^2 - \frac{b}{m} k_d s^2 - \frac{b}{m} k_p s - \frac{b}{m} k_i = 0$$

$$= s^3 + \left(\frac{q}{m} - \frac{b}{m} k_d\right) s^2 - \frac{b}{m} k_p s - \frac{b}{m} k_i = 0 \quad \text{--- (2)}$$

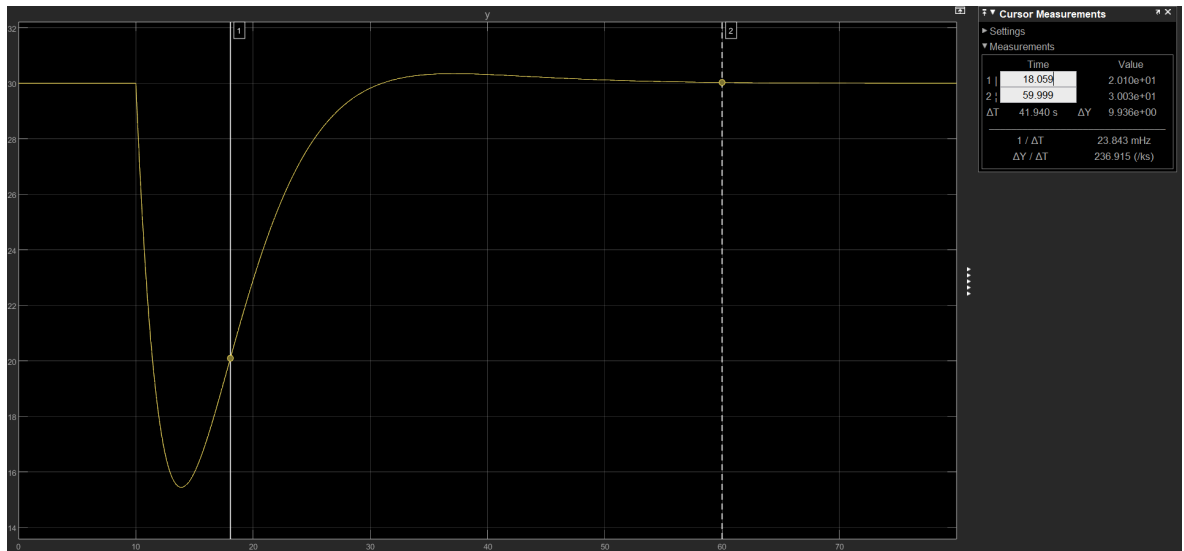
(1) & (2) we get.

$$s^2 = \frac{q}{m} - \frac{b}{m} k_d = 0.6$$

$$\Rightarrow \boxed{k_d = \frac{q - 0.6m}{b}}$$

$$s = -\frac{b}{m} k_p = -0.2 \Rightarrow \boxed{k_p = -\frac{0.12m}{b}}$$

$$\text{constant term} \Rightarrow -\frac{b}{m} k_i = 0.008 \Rightarrow \boxed{k_i = \frac{-0.008m}{b}}$$



- 8) Use Simulink and tune the controller by varying the pole placement, p , such that the relative distance between the vehicles is $d \geq 15$ meter for the implemented scenario:

Calculations for case a and b:

$$(s+p)^3 = s^3 + p^3 + 3ps^2 + 3p^2s \quad \text{--- (1)}$$

$$s^3 + \frac{a}{m}s^2 - \frac{b}{m}kds^2 - \frac{b}{m}kps - \frac{b}{m}ki \quad \text{--- (2)}$$

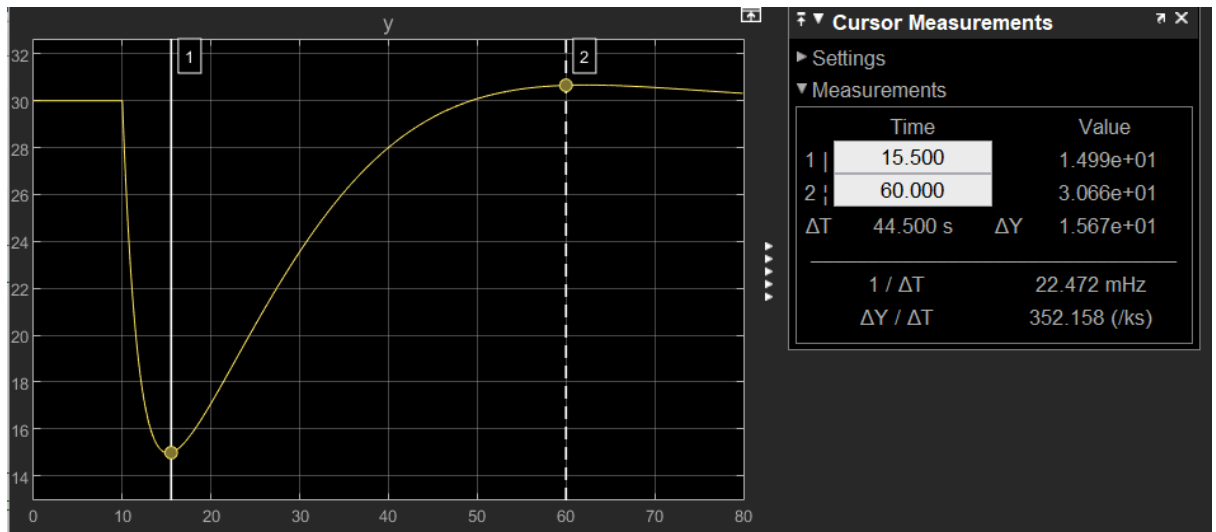
$$kd = \frac{-(3p^2xm + a)}{b}$$

$$kp = \frac{-3p^2xm}{b}$$

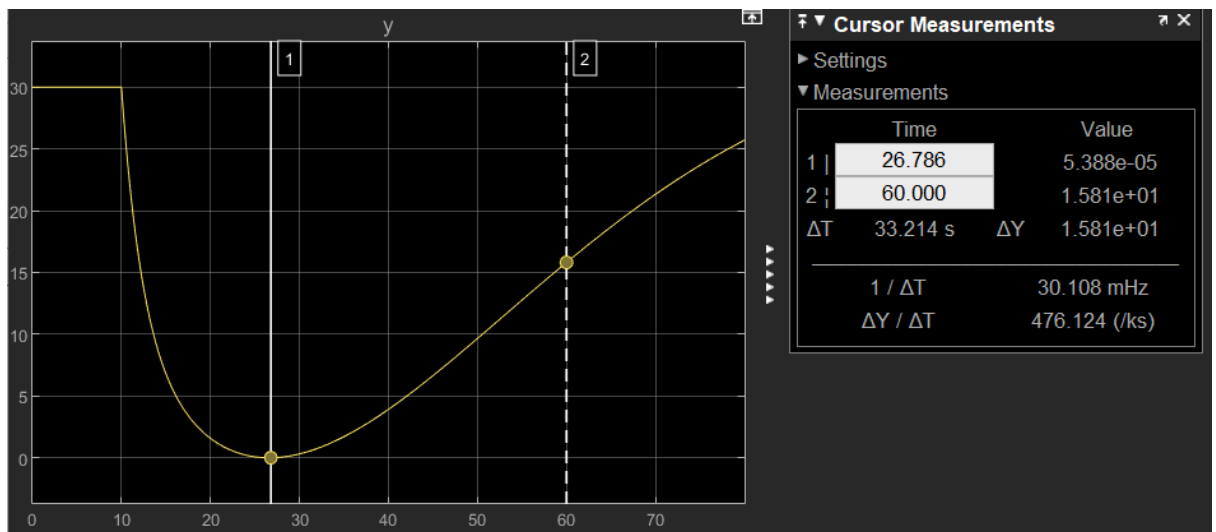
$$ki = \frac{-p^3xm}{b}$$

a) What pole placement gives $d = 15$ meter?

pole = 0.16045



b) What is the highest value of p which does not result in a collision, $d = 0$ meter?



pole = 0.09319

NOTE on problem 8:

We know it's a little bit weird that our pole is positive and still not unstable, as we discussed with one of the tutors 17/9-2024. He said that it's still working and assumed that we might have missed or made some wrong changes in the signs (- and +). But unless the results are stable we are good to go and it's working. If we need to redo something or have a quick look we will do it directly, just so this doesn't affect our grade because we know that it could be a miss somewhere.